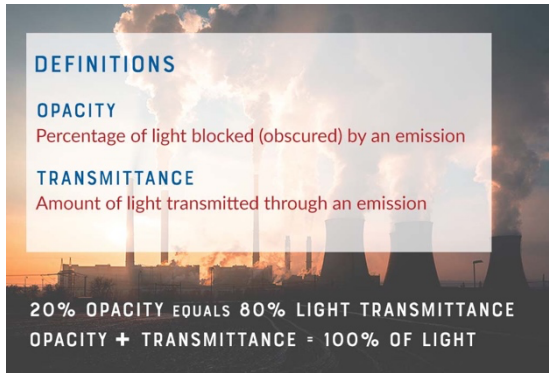
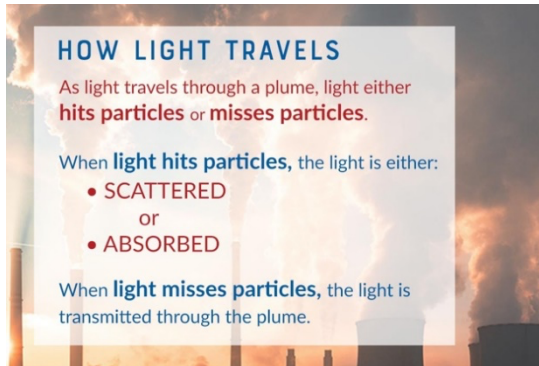
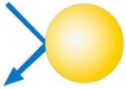
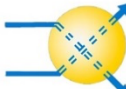
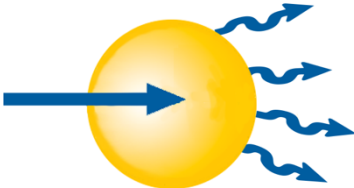


Script

Frame	VoiceOver / Spoken Word	Video /Action
1	The properties of light and how they affect visible emissions observations, particularly Method 9.	 1-cover-photo.jpg
2	We define opacity as the amount of background blocked by an emission. Opacity is the obscuring power of the plume - expressed as a percent. Transmittance is the amount of light that passes unaffected through an emission, also expressed as a percentage.	 2-definition-opacity-transmittance.jpg
3	As light passes through a plume, it either hits or misses particles. Generally, when light hits particles, it is scattered and/or absorbed. In limited circumstances, light can be transmitted through a particle. The vast majority of light that is transmitted through a plume misses particles.	 3-vocabulary-transmission.jpg Need animation here instead of a text slide
4	Scattered light is reflected, refracted, or a combination of both. Reflected light does not enter the particle. It is reflected like a mirror.	Should we get an animation here? yes

	In refraction, the light enters the particle and changes direction.	WHEN LIGHT IS SCATTERED When light hits particles and is SCATTERED, it is either reflected or refracted. REFLECTION  The light bounces off and does not enter the particle REFRACTION  The light enters the particle and changes direction 4-scattering.jpg I would animate and title Scattered or Maybe Light is Scattered by ---strike the red part and put small by in front of Reflection and Refraction ---- if using words we can minimize the busyness
5	A plume consisting of purely white particles scatters light.	 5-white-emission.jpg
6	A plume consisting of purely black particles absorbs the light, the light is re-emitted as infrared (heat) which we cannot see.	ABSORPTION Light enters the particle and is converted to heat  6-absorption.jpg Animation here.

7	How do the properties of light affect visible emission observations?	 7-properties-affect-observation.jpg
8	On a clear, bright day in full sun, the scattering effect of reflection and refraction is at its maximum with white smoke. Observers naturally will overstate the opacity of the plume, especially in opacity levels below 50%. Under these conditions, readings for black tend to be very accurate.	 8-reflection.jpg
9	On an overcast day, it can be difficult to read black smoke due to the lack of contrast between the color and luminescence of the emission and the background of gray or dark clouds. The natural tendency is to under-report the black emission. It is helpful to use a solid object in the background, such as a branch, as a reference when the clouds in the background are changing.	

10	A white smoke plume with 25% opacity will be more visible to the human eye than a black smoke plume with 25% opacity. It is important to not confuse the visibility of the plume with the visibility of the background. Remember you are estimating opacity based on the percent of the background that is obscured. This is why it is important to choose a background that is high in contrast and also high in luminosity. Light scattering mechanics apply not only to emissions, but the background as well.	 10-types-of-smoke.jpg
11	Smoke school instructors show students how to adjust according to current conditions during field certification training. With experience, observers learn to adjust to the varying sky conditions.	 11-plume-appearance.jpg
	CAA's online visible emission course reviews additional variables that can affect visible emissions observations.	 12-last-slide.jpg